

and do rule out order effects on accuracy based on the first two designs.

3.2.4 Participant Recruitment

Participants for the first and third designs, which constitute the majority, were recruited through teleradiology companies. The teleradiology companies allow us to recruit several experts in a relatively liquid spot market, a practice that is now common for decision experiments with non-expert subjects (Hunt and Scheetz, 2019). Most healthcare providers in the US rely on these companies’ services, although many large hospitals have on-call radiologists (Rosenkrantz et al., 2019). We work with teleradiology companies that serve US hospitals and offer the services of both US-based and non-US-based radiologists. Our contracts with teleradiology companies specify a piece-rate, and the companies, in turn, compensate the participants with a piece-rate.¹³ In addition, we provided monetary incentives for accuracy to a subset of radiologists, as described in the next section.

The second design required us to work with a partner who could guarantee subjects’ participation over several months. We collaborated with VinMac healthcare system in Vietnam to recruit their staff radiologists to ensure continued participation. VinMac is in the process of developing its own in-house AI capabilities and was willing to assist with our experiment in exchange for recognition in a publication of the resulting dataset. The VinMac radiologists did not receive any payments to participate in the experiment but we find that their performance is very close to the performance of the tele-radiologists.

In total, 180 radiologists participated in our experiment. Close to 17% of our participants are US-based, 38% have a degree from a US institution, 80% are affiliated with a large clinic, and 61% with an academic institution. As demonstrated in appendix C.3, the quality of the assessments made by the radiologists in our study is comparable to that of the staff radiologists from Stanford University Hospital, who originally diagnosed the patient case.

3.2.5 Incentives

We cross-randomize incentives for accuracy in the first and third designs but not the second because of the specific ways in which our partner’s radiologists are employed.¹⁴ Payments were determined following the binarized scoring rule in Hossain and Okui (2013), where truth

¹³We worked with three companies with piece-rates ranging from \$7.50 to \$13.00.

¹⁴There does not appear to be a consensus in the experimental literature on whether incentives are superior to non-incentivized responses when eliciting beliefs (see Danz, Vesterlund, and Wilson, 2020; Charness, Gneezy, and Rasocho, 2021). A comparison between the incentivized and non-incentivized participants allows us to answer this question in the context of this experiment.

is determined as described in section 3.3.1 below. This incentive scheme uses a loss function of the mean squared prediction error, averaging over patient cases and pathologies, and the respondents earn a fixed bonus of \$120 if a random draw is less than the loss function. This bonus is more than 20% of the base payment to teleradiology firms. We explain to the participants that expected payments are maximized if they provide their best estimates using a non-mathematical description of the payment rule. We specify the distribution so that 30% of pilot participants would earn the bonus, cross-randomized with the other two treatment arms.

3.3 Implementation and Data Collection

3.3.1 Patient Cases and Ground Truth

The experiment uses 324 historical patient cases with potential thoracic pathologies from Stanford University’s healthcare system. For each case, we have access to the chest X-ray and the clinical history in the form of the primary provider’s written notes, the patient vitals, and demographics.¹⁵ The use of retrospective cases allows us to avoid ethical and other issues that would arise when experimenting in high-stakes settings.

Our analysis requires constructing ω_i for each patient case. There are important challenges in using an observational dataset of patient health records to construct this field. One approach would be to use the results from further medical tests. Unfortunately, definitive gold-standard tests do not exist for most thoracic pathologies.¹⁶ Even when follow-up tests are conducted, they are selected on the likely presence of a pathology, an issue referred to as the selective labels problem (e.g. Mullainathan and Obermeyer, 2019). Medical outcomes from patient health records also do not suffice because actions taken by the treating physician in response to the radiology report contaminate these measures. Recent literature has suggested instrumental variables approaches for solving this selective labels problem, but this work targets population quantities and not a “ground truth” on each case (e.g. Chan et al., 2022; Mullainathan and Obermeyer, 2019).

We construct ω_i by aggregating the assessment of a group of expert radiologists, an approach common in computer science (Sheng et al., 2008; Mccluskey et al., 2021). Specifically, we

¹⁵All cases are first encounters with no prior X-ray as a comparison. We started with 500 cases that fit these primary criteria. We omitted pediatric cases from this set. Finally, a radiologist reviewed the cases to remove instances with poor image quality. The clinical history was manually reviewed to remove patient-identifiable information and cleared for public release.

¹⁶Many pathologies do not have commonly used non-imaging-based diagnostic tools. For instance, the presence of cardiomegaly – an enlarged heart – can only be determined using imaging tools, thoracic surgery, or an autopsy.

ask five board certified radiologists from Mount Sinai to read each of the 324 cases using the interface described above with the available X-ray and clinical history. For each case i and expert radiologist r , we, therefore, obtain $\pi_r(\omega_i = 1 | s_{i,r}^E)$ for each pathology, which we aggregate to generate ω_i . Specifically, we classify $\omega_i = 1$ if $\sum_r \pi_r(\omega_i = 1 | s_{i,r}^E) / 5 > 0.5$. This approach immediately addresses the selective labels problem because the availability of assessments is not selected on the likelihood of a pathology being present. Results in [Wallsten and Diederich \(2001\)](#) suggest that, under weak conditions that allow for measurement error in the reports and correlations across reports, the aggregate opinion of several experts is highly diagnostic as long as the experts are median unbiased. To assess robustness, we aggregate $\pi_r(\omega_i = 1 | s_{i,r}^E)$ using both a log-odds average and a leave-one-out mean of the assessments of all the radiologists in our experiment in the clinical history treatment condition.

3.3.2 Experimental Interface and Data Collected

We developed the interface to present the patient cases and to collect radiologists’ predictions and decisions in collaboration with board certified radiologists at Stanford University Hospital and Mt. Sinai Hospital. In contrast to free-text reports, we designed it to generate structured and quantitative data on chest X-ray responses that resemble a typical radiological report. A guiding principle in the design was to mimic clinical practice and to present and obtain all clinically relevant information. We briefly describe this interface and provide images and further details in appendix [B](#).

On the landing page of each case, a high-resolution image of a patient’s X-ray is presented to the radiologist, with the functionality to zoom and adjust brightness and contrast. When the experiment calls to show the clinical history, the interface presents clinical notes, vitals, and laboratory results available at the time the X-ray was originally ordered. If the experiment provides AI assistance, participants are shown AI predictions for fourteen pathologies and a composite prediction for whether or not there are any relevant findings.

The data entry interface collects a radiologist’s assessments of the probability that various thoracic pathologies are present for a given case. The probability that a pathology is present given the available information, i.e. $p_r(\omega = 1 | s)$, is elicited using a continuous slider. We visually subdivide possible responses into five intervals with standard language labels used in written radiological reports to aid the participants.¹⁷ Our radiology collaborators grouped pathologies into eight exclusive parent categories based on their type. Each group has children

¹⁷The specific labels are “Not present”, “Very Likely”, “Unlikely”, “Possible”, “Likely”, and “Highly Likely”. Several radiological publications have suggested such standardized language for radiological reports. See for instance [Panicek and Hricak \(2016\)](#).

that are more specific, which may be further subdivided in some cases. The groups all correspond to a standard class of pathologies. For instance, “airspace opacity” is distinct from a “cardiomediastinal abnormality.” In the main text we focus on the parent categories with AI predictions and drop further subdivisions from the analysis. We refer to those as top-level pathologies. Our results are robust to including the lower-level pathologies in the analysis as we show in appendix C.4. The interface categorizes thoracic pathologies into groups by type to ease data entry. For example, allowing the user to simultaneously set the assessed probability of each disease in a specific category to zero.¹⁸ In addition, we elicited an overall bottom-line assessment of whether the radiologists considers the case normal or not.

We also ask for a binary “treatment/follow-up” recommendation for each pathology that is not definitively ruled out.¹⁹ We will interpret this input as $a_r^*(s)$. In a real clinical setting, a recommendation to follow-up could trigger the treating physician to prescribe additional medical tests or interventions with potential costs and benefits. Thus, an optimal recommendation trades off the cost of false positives and false negatives when recommending an action as in section 2. The probabilistic assessments with the follow-up decision will allow us to estimate radiologists’ relative cost of false positives and false negatives.

In addition to $p_r(\omega = 1 | s)$ and $a_r^*(s)$, we record active time, response times, and any click-stream data that results from the interaction with the interface. The participants are not explicitly informed about this monitoring, and there are no explicit time limits. Our experiment runs remotely, and participants connect to a server, which hosts the interface and records responses. The interface has been extensively tested beforehand on different browsers by conducting pilots with every company we recruited the participants from.²⁰

¹⁸Our radiology collaborators grouped pathologies into eight exclusive parent categories based on their type. Each group has children that are more specific, which may be further subdivided in some cases. The groups all correspond to a standard class of pathologies. For instance, an “airspace opacity” is distinct from a “cardiomediastinal abnormality.” The more specific disease of “bacterial pneumonia” manifests as an airspace opacity. This structure reduced the burden on our participants, and we piloted the interface with several radiologists specializing in the interpretation of chest X-rays. Prior clinical research on AI in chest X-Ray image classification has used similar hierarchies (see [Seah et al., 2021](#), for example).

¹⁹The binary treatment/follow-up decision is only asked for pathologies where a follow-up is clinically relevant. This includes all pathologies with AI assistance.

²⁰This interface is browser-based and built using the o-tree framework [Chen et al. \(2016\)](#). Since we are not directly communicating with our participants we also deploy a device fingerprinting service from [fingerprint.com](#) to ensure that there are no repeat participants.

3.3.3 Participant Training

We train the participants using a combination of written instructions and a video. The materials provide an overview of the experimental tasks, the interface, and information about the AI assistance tool. The firms and the participants know that the research study involves retrospective patient cases. To train participants on the AI tool, we provide them with materials that explain the development of the algorithm, present metrics of its performance on various diseases, and summarize the algorithm’s performance relative to radiologists based on prior research. The participants are informed that the algorithm only uses the chest X-ray to form predictions, and this knowledge is later tested in a comprehension question. In addition, we show the participants fifty example cases that show the X-ray and clinical history next to the AI output. After the instructions, they answer eight comprehension questions, which the participants must answer correctly before proceeding to the experiment. We also include an endline survey. We do not directly interact with the subjects except to field questions about the experiment or provide tech support.²¹ The complete set of instructions is provided in appendix B.2.

4 Estimated Treatment Effects

4.1 Overall Performance of AI and Radiologists

This section’s analysis focuses on measures of performance (deviation from ground truth, incorrect decision), deviation from AI prediction, and measures of effort. Table 1 summarizes the data on these measures and sample sizes from our experiment. The main text focuses on top-level pathologies with AI with robustness to other pathology groups relegated to appendix C.4.²²

Radiologists make the correct follow-up/treatment recommendation on approximately 70% of case-pathologies on average and spend ~2.8 minutes per case with large variability across cases. All summary statistics are very similar across the three experimental designs (tables C.7 and C.9). For instance, the average deviation from the ground truth for the three designs ranges from 0.191 to 0.232, and average active time ranges from 2.58 to 3.03 minutes.

²¹We are blinded to the participants’ treatment status while the experiment is in progress.

²²These pathology groups and, unless otherwise noted, the subsequent analyses were pre-registered (see SSR Registration 9620).

Table 1: Summary Statistics

	All Designs		Design 1	
	Mean	SD	Mean	SD
	(1)	(2)	(3)	(4)
Reported Probability	0.233	0.290	0.214	0.287
Decision	0.322	0.467	0.277	0.448
Deviation from Ground Truth	0.223	0.281	0.218	0.290
Deviation from AI	0.191	0.169	0.201	0.171
Correct Decision	0.695	0.460	0.736	0.441
Active time	2.82	2.63	3.03	3.17
Observations	36,280		13,440	
Radiologists	180		112	

Note: This table presents summary statistics of the experimental data. Columns (1) and (2) present the mean and standard deviation for all designs while Columns (3) and (4) present the same for design 1 only. Decision is an indicator for whether treatment/follow-up is recommended, correct decision is an indicator for whether the decision matches the ground truth, deviation from ground truth is the absolute difference between the reported probability and the ground truth, deviation from AI is the absolute difference between the expert’s reported probability and the AI’s reported probability, active time is measured in minutes.

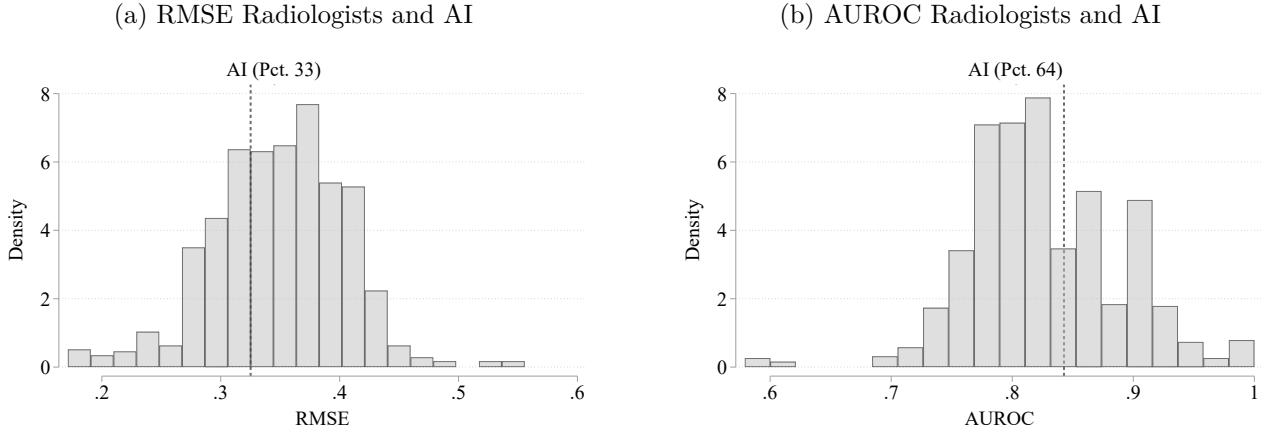
In a comparison study with three radiologists, [Irvin et al. \(2019\)](#) show that the CheXpert model yields a better classifier than two out of three radiologists on five pathologies and all three on a subset of three pathologies. Our results may differ from that because we use a different pool of radiologists, a different sample of cases, and reads with contextual information (clinical history) to construct the ground truth. The latter two differences raise the bar for the AI because they reflect differences in the data-generating process.

To benchmark the quality of AI predictions in our sample and the radiologists in our experiment, we compare our participant pool with the AI input using two performance measures. The first measure is derived from the receiver operating characteristic (ROC) curve, which measures the trade-off between the false positive and the true positive rate of a classifier as the cutoff for classifying a case as positive or negative is varied. It uses only ordinal information about the AI. A classifier that is not better than random has an AUROC value of 0.5 whereas a perfectly predictive classifier has a value of one. The second measure is the root mean squared error (RMSE), that utilizes cardinal information about the AI prediction. A

lower RMSE indicates higher performance.²³ We pool the data for top-level pathologies with AI for each radiologist’s reports and for the AI’s prediction.

Figure 1 shows significant heterogeneity in performance across radiologists as well as the scope for AI assistance to improve radiologist performance. The AUROC shows that the bottom tail of the distribution of radiologists performs worse than the AI, whereas the upper tail predicts close to perfectly. This heterogeneity aligns with findings from observational data based on only treatment decisions (see Chan et al., 2022). The AI is more predictive than approximately two-thirds of radiologists, whether compared using AUROC or RMSE. Thus, there is ample room for AI assistance to improve the performance of radiologists. In fact, a majority of radiologists would do better on average by simply following the AI prediction. We present pathology-specific comparison results in appendix C.2.

Figure 1: Comparing AI performance to radiologists



Note: These histograms show distributions of two different accuracy measures of radiologist assessments alongside the AI’s accuracy. The left graph shows the distribution of the RMSE while the right shows the distribution of the average AUROC. Both distributions are shrunk to the grand mean using empirical Bayes. These measures are for each radiologist and include the top-level pathologies. The dotted line is the average measure of the AI algorithm for the corresponding distribution. Only the assessments where contextual history information is available for the radiologists but not the AI prediction are considered. Robustness by design and ground truth definition can be found in sections C.4.1 & C.4.2.

We also compare the performances of our participants and the radiologist who originally diagnosed each patient case in appendix C.3.²⁴ There is no discernible difference between

²³An important distinction between the two measures is that the AUROC is ordinal whereas RMSE is cardinal.

²⁴We classified the original free text radiology reports associated with each case as positive, negative, or uncertain for each pathology using the CheXbert algorithm described in Smit et al. (2020). To facilitate comparisons, we also discretized the probability assessments elicited during the experiment into these three categories. Then, we compared the accuracy of the original reads against the radiologists participating in the experiment.